Decoding continuous sleep states in mice from regional fMRI dynamics

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Introduction

Human fMRI studies have validated sleep-stage classification across subjects,[1] while mouse fMRI studies have predicted state transitions.[2] We tested continuous sleep-state reconstruction in held-out mice, its dependence on temporal decoding, and regional contributions.

Methods

We analyzed 45 sessions (24 mice; 180 hours; 324,000 volumes) with electrophysiological Awake, non-rapid-eye-movement (NREM) and rapid-eye-movement (REM) labels. Five features (amplitude, difference, mean, standard deviation and slope) from 54 anatomical regions of interest (ROIs) provided 270 inputs to elastic-net multinomial logistic regression (Fig. 1A); selected centered windows spanned 21 TR (TR = 2 s). Nested leave-one-mouse-out validation separated training, tuning and testing. A hidden Markov model (HMM) decoded full sessions offline. Comparators were single-volume ROI classification, linear/nonlinear motion-only classifiers (logistic regression/gradient boosting), and regional classification after training-fitted motion regression (Table 1). ROI importance used held-out F1 decreases over 20 joint circular shifts of each ROI's five features. Five-feature L2 norms summarized standardized coefficients relative to the other states. Mouse bootstraps quantified uncertainty.

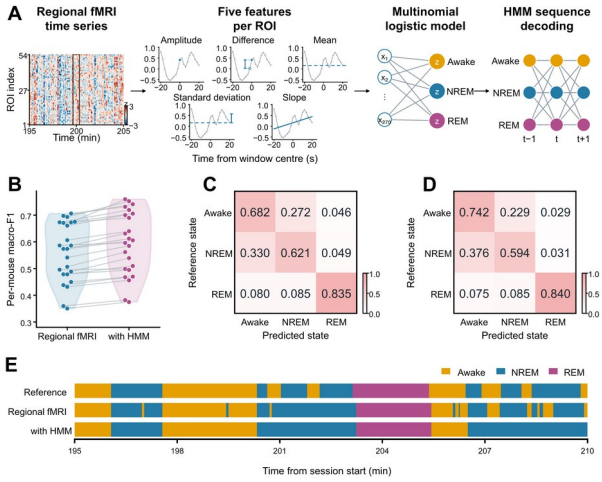


Figure 1. Sleep-state decoding. A, Regional feature extraction, classification and hidden Markov model (HMM) decoding. B, Mouse-level macro-F1 (n = 24): points are mice, lines connect paired scores, violins show distributions. C,D, Regional/HMM confusion matrices: rows are reference states, columns predictions; values are row-normalized. E, A post hoc high-performing 15-min sequence; gold/blue/purple indicate Awake/NREM/REM.

Results

HMM decoding increased pooled macro-F1 from 0.605 to 0.651 (Table 1). All 24 mice improved: mean paired increase, 0.034 (95% CI, 0.028–0.042; Fig. 1B). Awake recall increased from 0.682 to 0.742, whereas NREM recall decreased from 0.621 to 0.594. REM recall changed from 0.835 to 0.840 and precision from 0.363 to 0.477 (Fig. 1C,D); REM constituted 3.11% of volumes.

Table 1. Pooled held-out performance (24 mice).

Model	F1	95% CI
Single-volume fMRI	0.428	0.394–0.461
Regional fMRI	0.605	0.566–0.637
Regional fMRI + HMM	0.651	0.611–0.682
Motion only: linear	0.477	0.451–0.498
Motion only: nonlinear	0.536	0.506–0.561
Motion-regressed fMRI	0.590	0.549–0.623

F1 = pooled macro-F1. CIs: mouse bootstrap (2,000 resamples for regional and HMM; 10,000 otherwise).

The dentate gyrus (DG) showed the largest F1 losses for Awake (5.61 percentage points) and NREM (5.69), followed by ventral retrosplenial cortex (RSPv) and somatosensory barrel field (Fig. 2A). For REM, ventral retrosplenial cortex ranked first (30.02 points; descriptive 95% CI, 24.06–36.49), followed by ventral anterior cingulate cortex (5.17) and presubiculum (5.13). Ventral retrosplenial cortex also had the largest mean REM coefficient norm and ranked among the top ten in all 24 models (Fig. 2B); its REM F1 loss remained 29.20 points after motion regression. The motion-regressed versus nonlinear motion-only paired macro-F1 difference was 0.028 (95% CI, −0.001–0.060).

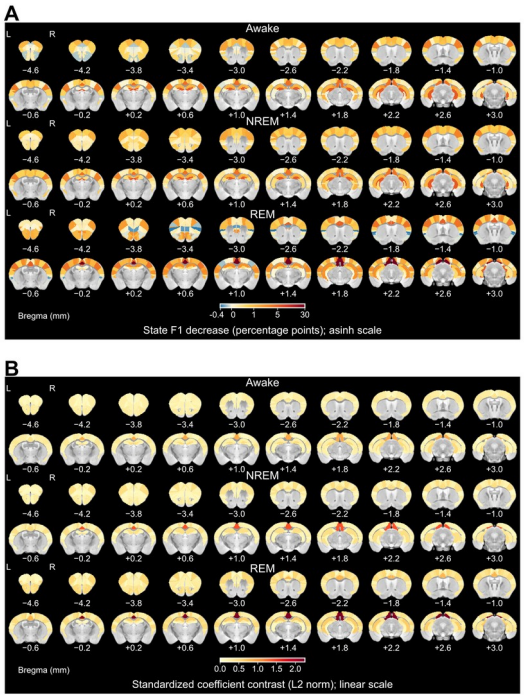


Figure 2. State-specific ROI contributions. A, Mean classifier F1 loss after jointly shifting each ROI's five features (Awake/NREM: 24 mice; REM: 22); asinh scale, ticks in percentage points. B, Mean five-feature L2 norm of standardized coefficient contrasts against the mean of the other two states (24 models; linear scale). Colors show values within modeled gray matter. Each state has 20 coronal sections; nominal Bregma coordinates (mm) follow the supplied reference.

Discussion

DG importance for Awake/NREM discrimination is consistent with NREM-associated infraslow neuronal activity reported in mouse DG.[3] RSPv dominance for REM is consistent with mouse studies linking retrosplenial cortical activity to REM substage progression.[4] Together, these findings suggest that regional fMRI features capture state-dependent hippocampal and cortical dynamics.

Conclusion

Regional fMRI dynamics supported continuous sleep-state reconstruction in held-out mice within this dataset. Independent recordings and further motion controls are needed to assess transfer to other acquisitions and use alongside electrophysiological scoring.

References

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